Variable:-

In Python, a variable is a symbolic name that is used to refer to a value or an object in memory. Python variables are versatile and can be used to store a wide range of data types. Here's a more detailed look at Python variables:

Variables do not need to be declared with any particular *type.*

Variable Names: -

- A variable name must start with a **letter or the underscore** character**,**
- A variable name cannot start with a number
- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
- Variable names are case-sensitive (age, Age and AGE are three different variables)

Camel Case

```
myVariableName = "John"
```

Pascal Case

```
MyVariableName = "John"
```

With underscore

```
My_Variable_Name = "John"
```

Assign multiple values to variable

```
x, y, z = "Orange", "Banana", "Cherry"
print(x)
print(y)
print(z)
```

```
Orange
Banana
Cherry
```

Python Casting

There may be times when you **want to specify a type on to a variable.** This can be done with **casting.** Python is an object-orientated language, and as such it uses classes to define data types, including its primitive types.

Casting in Python is therefore done using constructor functions:

int() - constructs an integer number from an integer literal, a float literal (by removing all decimals), or a string literal (providing the string represents a whole number)

float() - constructs a float number from an integer literal, a float literal or a string literal (providing the string represents a float or an integer)

str() - constructs a string from a wide variety of data types, including strings, integer literals and float literals

Casting to Integers	*Casting to float*	*Casting to string*

```
x = int(1)   # x will be 1
y = int(2.8) # y will be 2
z = int("3") # z will be 3

print(type(x))
print(x)
print(type(y))
print(y)
print(type(z))
print(z)
```

```
<class 'int'>
1
<class 'int'>
2
<class 'int'>
3
```

```
x = float(1)     # x will be 1.0
y = float(2.8)   # y will be 2.8
z = float("3")   # z will be 3.0
w = float("4.2") # w will be 4.2

print(type(x))
print(x)
print(type(y))
print(y)
print(type(z))
print(z)
print(type(w))
print(w)
```

```
<class 'float'>
1.0
<class 'float'>
2.8
<class 'float'>
3.0
<class 'float'>
4.2
```

```
x = str("s1") # x will be 's1'
y = str(2)    # y will be '2'
z = str(3.0)  # z will be '3.0'

print(type(x))
print(x)
print(type(y))
print(y)
print(type(z))
print(z)
```

```
<class 'str'>
s1
<class 'str'>
2
<class 'str'>
3.0
```